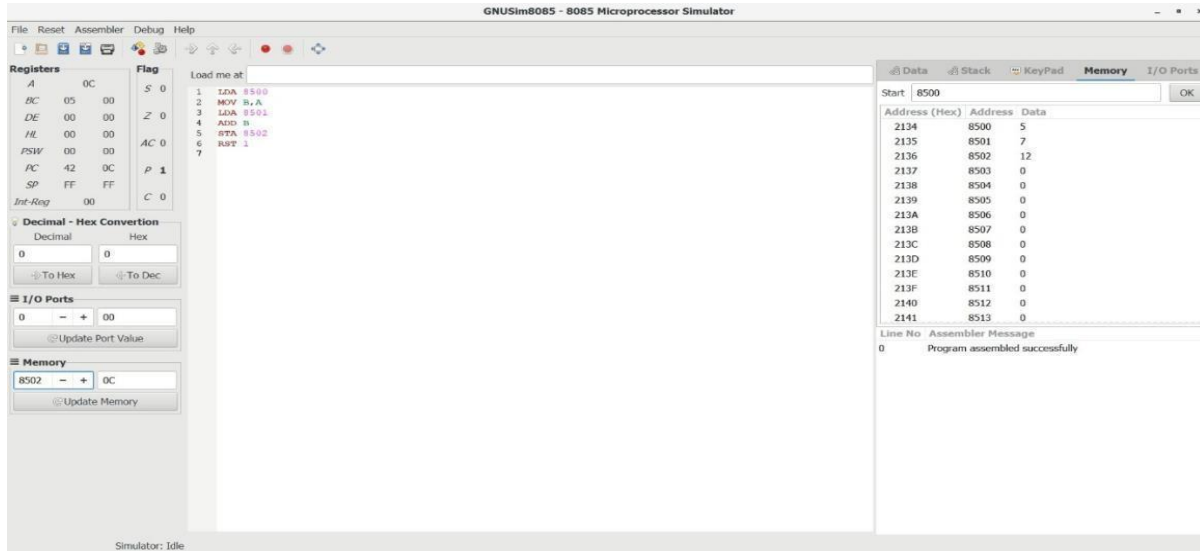


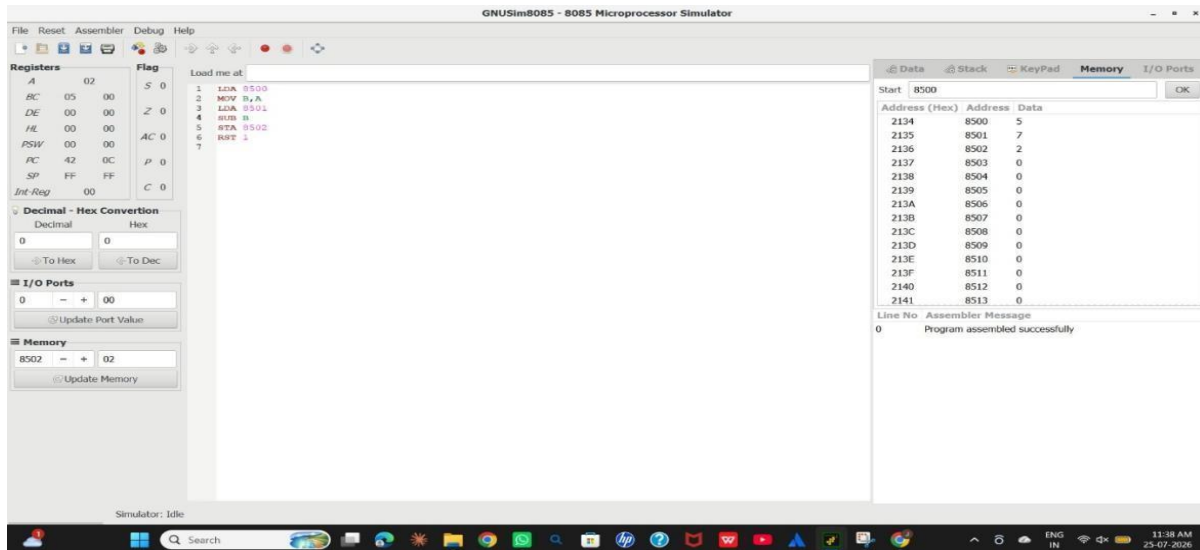
CSA1204 COMPUTER ARCHITECTURE

8085 LAB PROGRAMS

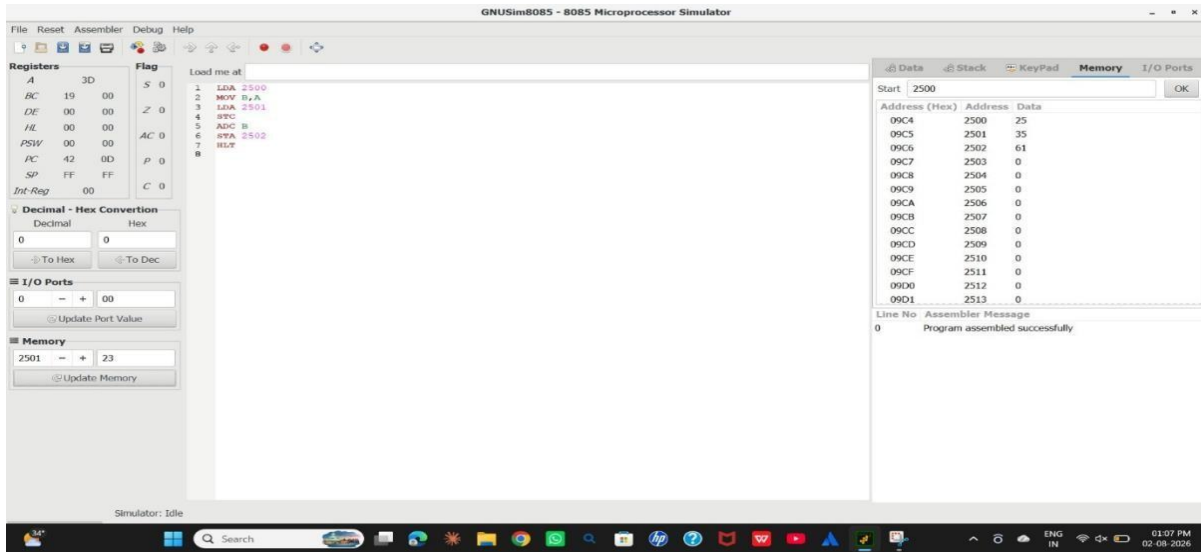
Q1. Addition of 2 Numbers



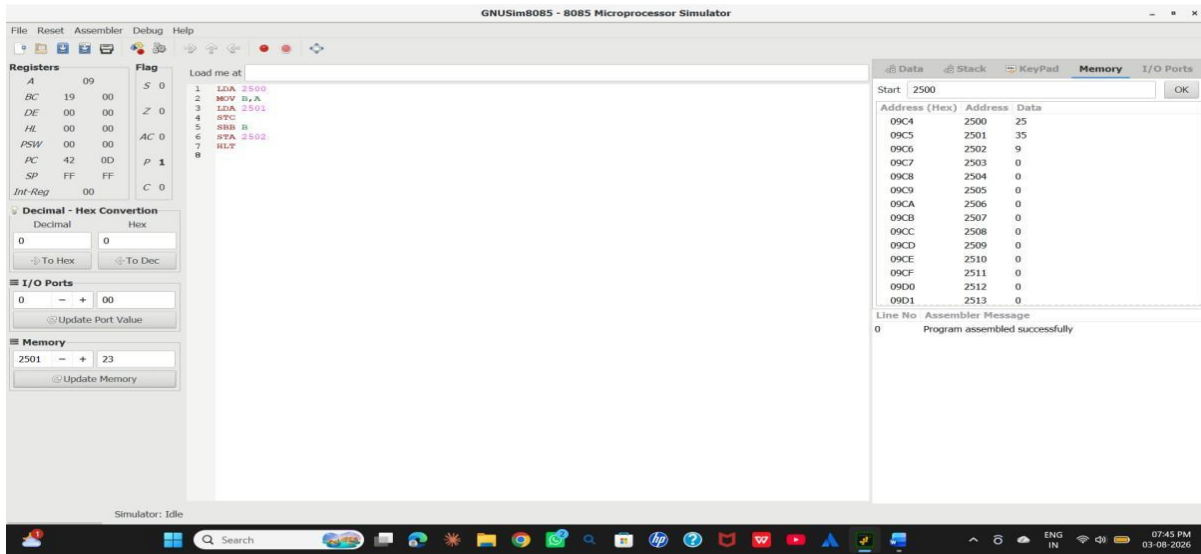
Q2. Subtraction of 2 Numbers



Q3. Addition With Carry (ADC)



Q4. Subtraction With Borrow (SBB)



Q5. Multiplication of 2 Numbers

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers: A 32, BC 0A 00, DE 00 00, HL 00 00, PSW 00 00, PC 42 13, SP FF FF, Int-Reg 00. Flag: S 0, Z 1, AC 0, P 1, C 0.

Decimal - Hex Conversion: Decimal 0, Hex 0. Buttons: To Hex, To Dec.

I/O Ports: 0, Update Port Value.

Memory: 2501, Update Memory.

Load me at:

```
1 LDA 2500
2 MOV B,A
3 LXA 2501
4 MOV C,A
5 MVI A,00
6 LOOP: ADD B
7 DCR C
8 JNZ LOOP
9 STA 2502
10 HLT
```

Start: 2500. OK

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	5
09C6	2502	50
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No. Assembler Message
0 Program assembled successfully

Simulator: Idle

30°C Mostly cloudy

Search

ENG IN

13:31 28-07-2026

Q6. Division of Two Numbers

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers: A 05, BC 0A 02, DE 05 00, HL 00 00, PSW 00 00, PC 42 1C, SP FF FF, Int-Reg 00. Flag: S 0, Z 1, AC 1, P 1, C 1.

Decimal - Hex Conversion: Decimal 0, Hex 0. Buttons: To Hex, To Dec.

I/O Ports: 0, Update Port Value.

Memory: 2501, Update Memory.

Load me at:

```
1 LDA 2500
2 MOV B,A
3 LXA 2501
4 MOV C,A
5 MVI D,00
6 MOV A,B
7 LOOP: RRB C
8 JC STOP
9 INR D
10 JND LOOP
11 STOP: ADD C
12 STA 2503
13 MOV A,D
14 STA 2502
15 HLT
```

Start: 2500. OK

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	2
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No. Assembler Message
0 Program assembled successfully

Simulator: Idle

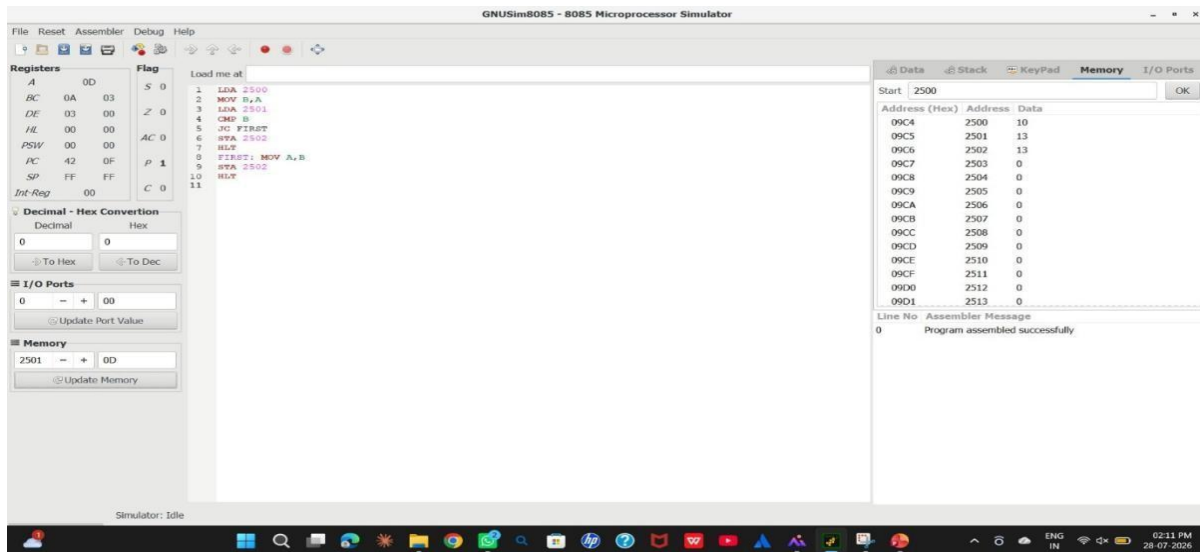
30°C Mostly cloudy

Search

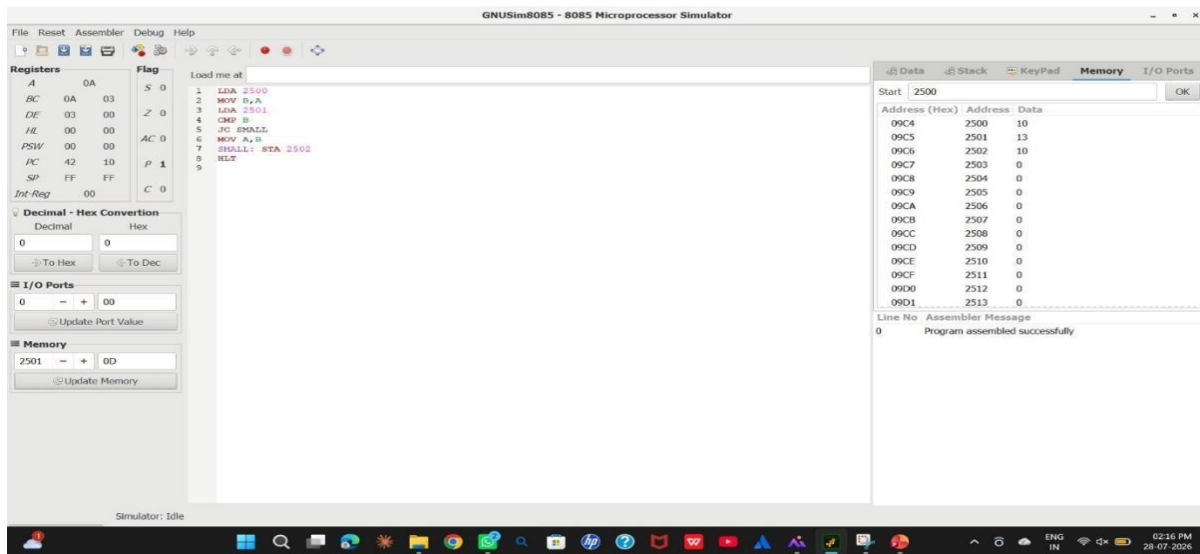
ENG IN

01:42 PM 28-07-2026

Q7. Greatest of 2 Numbers



Q8. Smallest of 2 Numbers



Q9. Factorial of a Number

Flag

S 0

Z 1

DD AC 0

00 P 1

FF C 0

Conversion

Hex

0

< To Dec

00

Port Value

05

Memory

Load me at

```
1 LDA 2500
2 MOV C,A
3 MVI B,01H
4
5 FACT: MOV A,C
6 CPI 00H
7 JZ DONE
8
9 MOV D,C
10 MOV A,B
11 MOV E,A
12 MVI A,00H
13
14 MUL: ADD E
15 DCR D
16 JNZ MUL
17
18 MOV B,A
19 DCR C
20 JMP FACT
21
22 DONE: MOV A,B
23 STA 2501
24 HLT
```

Data

Stack

KeyPad

Me

Start 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	120
09C6	2502	2
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Q10. Packed BCD to Binary Conversion

S 0

Z 0

AC 1

P 1

C 0

Conversion

Hex

< To Dec

Value

Memory

```
1 LDA 2500
2 MOV B,A
3 ANI 0FH
4 MOV E,A
5
6 MOV A,B
7 ANI 0F0H
8 RRC
9 RRC
10 RRC
11 RRC
12 MOV D,A
13
14 MVI C,0AH
15 MVI A,00H
16
17 LOOP: ADD D
18 DCR C
19 JNZ LOOP
20
21 ADD E
22 STA 2501
23 HLT
```

Start 2500

Address (Hex)	Address	Data
09C4	2500	45
09C5	2501	33
09C6	2502	2
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Q11. GCD (HCF) of Two Numbers

The screenshot shows an 8085 assembly program for finding the GCD of two numbers. The program is loaded at address 02. The assembly code is as follows:

```
02 02 S 0 1 LDA 2500
18 18 Z 1 2 MOV B,A
09 DD AC 0 3 LDA 2501
00 00 P 1 4 MOV C,A
FF FF C 0 5
10 10 6 LOOP: MOV A,B
11 11 7 CMP C
12 12 8 JZ DONE
13 13 9 JC SECOND
14 14 10 SUB C
15 15 11 MOV B,A
16 16 12 JMP LOOP
17 17 13
18 18 14 SECOND: MOV A,C
19 19 15 SUB B
20 20 16 MOV C,A
21 21 17 JMP LOOP
22 22 18
23 23 19 DONE: MOV A,B
24 24 20 STA 2502
25 25 21 HLT
```

The program uses the following registers and flags:

- Registers: S, Z, DD, AC, P, FF, C
- Flags: S, Z, DD, AC, P, FF, C

The program is assembled successfully. The memory dump shows the following data:

Address (Hex)	Address	Data
09C4	2500	6
09C5	2501	8
09C6	2502	2
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message shows: 0 Program assembled successfully.

Q12. LCM of Two Numbers

The screenshot shows an 8085 assembly program for finding the LCM of two numbers. The program is loaded at address 08. The assembly code is as follows:

```
08 08 S 0 1 LDA 2500
18 18 Z 1 2 MOV B,A
DD DD 3 LDA 2501
00 00 AC 0 4 MOV C,A
23 23 P 1 5
FF FF C 0 6 MOV D,B
10 10 7 MOV E,C
11 11 8
12 12 9 LOOP: MOV A,D
13 13 10 CMP E
14 14 11 JZ DONE
15 15 12 JC ADDD
16 16 13 MOV A,E
17 17 14 ADD C
18 18 15 MOV E,A
19 19 16 JMP LOOP
20 20 17
21 21 18 ADDD: MOV A,D
22 22 19 ADD B
23 23 20 MOV D,A
24 24 21 JMP LOOP
25 25 22 DONE: MOV A,D
26 26 23 STA 2502
27 27 24 HLT
```

The program uses the following registers and flags:

- Registers: S, Z, DD, AC, P, FF, C
- Flags: S, Z, DD, AC, P, FF, C

The program is assembled successfully. The memory dump shows the following data:

Address (Hex)	Address	Data
09C4	2500	6
09C5	2501	8
09C6	2502	24
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message shows: 0 Program assembled successfully.

Q13. Smallest Number in an Array

Load me at

```
1 LXI H,2501
2 MOV B,M
3 MOV A,M
4 MOV D,A
5 DCR B
6
7 LOOP: INX H
8 MOV A,M
9 CMP D
10 JNC SKIP
11 MOV D,A
12
13 SKIP: DCR B
14 JNZ LOOP
15
16 MOV A,D
17 STA 2600
18 HLT
```

Data Stack KeyPad Memory

Start 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	25
09C6	2502	10
09C7	2503	45
09C8	2504	15
09C9	2505	30
09CA	2506	5
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Q14. Largest Number in an Array

Flag

S 0

Z 1

D AC 0

P 1

F C 1

Inversion

Hex

To Dec

0

t Value

E

emory

Load me at

```
1 LXI H,2501
2 MOV B,M
3 MOV A,M
4 MOV D,A
5 DCR B
6
7 LOOP: INX H
8 MOV A,M
9 CMP D
10 JC SKIP
11 MOV D,A
12
13 SKIP: DCR B
14 JNZ LOOP
15
16 MOV A,D
17 STA 2506
18 HLT
```

Data Stack KeyPad Memory

Start 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	25
09C6	2502	10
09C7	2503	45
09C8	2504	15
09C9	2505	30
09CA	2506	45
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Q15. Bubble Sort – Descending Order

```
1 LDA 2500
2 DCR A
3 MOV B,A
4
5 OUTER: LXI H,2501
6 MOV C,B
7
8 INNER: MOV A,M
9 MOV D,A
10 INX H
11 MOV A,M
12 MOV E,A
13 MOV A,D
14 CMP E
15 JNC NOSWAP
16
17 MOV M,D
18 DCX H
19 MOV M,E
20 INX H
21
22 NOSWAP: DCR C
23 JNZ INNER
24 DCR B
25 JNZ OUTER
26
27 HLT
```

Start 2500

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	8
09C6	2502	8
09C7	2503	2
09C8	2504	1
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Q16. Bubble Sort – Ascending Order

Flag

S 0

Z 1

AC 0

P 1

C 0

version

Hex

To Dec

value

ory

```
1 LDA 2500
2 DCR A
3 MOV B,A
4
5 OUTER: LXI H,2501
6 MOV C,B
7
8 INNER: MOV A,M
9 MOV D,A
10 INX H
11 MOV A,M
12 MOV E,A
13 MOV A,D
14 CMP E
15 JC NOSWAP
16 JZ NOSWAP
17
18 MOV M,D
19 DCX H
20 MOV M,E
21 INX H
22
23 NOSWAP: DCR C
24 JNZ INNER
25 DCR B
26 JNZ OUTER
27
28 HLT
```

Start 2500

Address (Hex)	Address	Data
09C4	2500	4
09C5	2501	1
09C6	2502	2
09C7	2503	5
09C8	2504	8
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Q17. Check Whether a Number Is Positive

The screenshot shows an 8086 assembly simulator. The assembly code is as follows:

```
1 LDA 2500
2 ANI 80H
3
4 JZ POSITIVE
5
6 MVI A, 01H
7 STA 2501
8 HLT
9
10 POSITIVE: MVI A, 00H
11 STA 2501
12
13 HLT
```

The flag register (F) shows: S 0, Z 1, AC 1, P 1, C 0. The memory window (Memory) shows the following data:

Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	0
09C6	2502	0
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message window shows: 0 Program assembled successfully.

Q18. Check Whether a Number Is Odd or Even

The screenshot shows an 8086 assembly simulator. The assembly code is as follows:

```
1 LDA 2500
2 ANI 01H
3
4 JZ EVEN
5
6 MVI A, 01H
7 STA 2501
8 HLT
9
10 EVEN: MVI A, 00H
11 STA 2501
12
13 HLT
```

The flag register (F) shows: S 0, Z 0, AC 1, P 0, C 0. The memory window (Memory) shows the following data:

Address (Hex)	Address	Data
09C4	2500	9
09C5	2501	1
09C6	2502	0
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message window shows: 0 Program assembled successfully.

Q19. 2's Complement of a Number

The screenshot shows an assembler interface with the following components:

- Registers:** A table on the left shows the status of various registers (FB, S, Z, AC, P, C).
- Assembly Code:** A central window displays the following code:

```
1 LDA 2500
2 CMA
3 INR A
4 STA 2501
5 HLT
```
- Hex Conversion:** A panel on the left includes a 'Decimal - Hex Conversion' section with a 'To Hex' button and a 'Hex' input field containing '0'. Below it is a 'Ports' section with a 'Try' input field containing '05'.
- Memory Map:** A table on the right shows memory addresses from 09C4 to 09D1. Address 2500 contains the value 5, and address 2501 contains the value 251. All other addresses contain 0.
- Assembler Messages:** A bottom panel shows a message: '0 Program assembled successfully'.

Q20. 1's Complement of a Number

The screenshot shows an assembler interface with the following components:

- Assembly Code:** A central window displays the following code:

```
1 LDA 2500
2 CMA
3 STA 2501
4 HLT
```
- Memory Map:** A table on the right shows memory addresses from 09C4 to 09D1. Address 2500 contains the value 55, and address 2501 contains the value 200. All other addresses contain 0.
- Assembler Messages:** A bottom panel shows a message: '0 Program assembled successfully'.

Q21. Swap Two Numbers

S 1

Z 0

AC 0

P 1

C 1

tion

2X

Dec

e

1 LDA 2500

2 MOV B,A

3

4 LDA 2501

5 MOV C,A

6

7 MOV A,B

8 STA 2501

9

10 MOV A,C

11 STA 2500

12

13 HLT

Start 2500

Address (Hex)	Address	Data
09C4	2500	15
09C5	2501	25
09C6	2502	0
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No

Assembler Message

0

Program assembled successfully